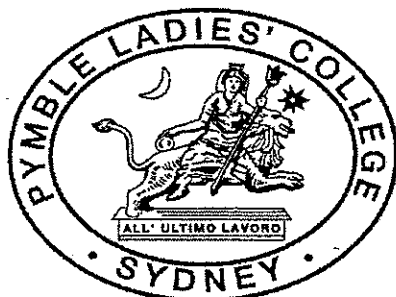


Mrs Collett
Ms Lau

Name:

Teacher:



2011

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 120

- Attempt Questions 1–8
- All questions are of equal value

Mark		/120
Rank		/18
Highest Mark		/120

Blank Page

Total marks – 120

Attempt Questions 1–8

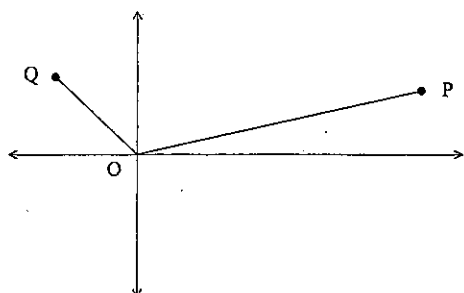
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)	Use a SEPARATE writing booklet	Marks
(a)	If $z = 3 + 2i$ and $\omega = 1 + i$, find in the form $a + ib$ where a and b are real	
(i)	$2z - i\omega$.	1
(ii)	$z\bar{\omega}$.	1
(iii)	$\frac{3}{\omega}$.	1
(b)	On the Argand diagram, sketch the locus of z described by the inequality $ z - 2 - 3i \geq z + i $.	2
(c)	Let $\alpha = -\sqrt{3} + i$.	
(i)	Express α in modulus-argument form.	2
(ii)	Show that α is a root of the equation $z^6 + 64 = 0$.	1
(iii)	Hence, find a real quadratic factor of the polynomial $P(z) = z^6 + 64$.	2

Question 1 continues on page 3

- (d) The diagram shows a complex plane with origin O.



Points P and Q represent non-zero complex numbers z and w respectively.

- | | | |
|-------|--|---|
| (i) | Write down the length of PQ in terms of z and w . | 1 |
| (ii) | Copy the diagram into your booklet. Construct point R that represents $z + w$.
What type of quadrilateral is OPQR? | 2 |
| (iii) | Prove that if $ z + w = z - w $, the complex number $\frac{w}{z}$ is imaginary. | 2 |

End of Question 1

Question 2 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) (i) Find the value of a and b such that

2

$$\frac{1}{(x-1)(2x+3)} = \frac{a}{x-1} + \frac{b}{2x+3}.$$

- (ii) Hence find $\int \frac{dx}{(x-1)(2x+3)}.$

2

- (b) Use the substitution $t = \tan x$ to find $\int \operatorname{cosec} 2x \, dx.$

3

- (c) Evaluate $\int_{-\frac{1}{2}}^0 \frac{dx}{2+4x+4x^2}.$

2

- (d) Evaluate $\int x(3^x) \, dx.$

3

- (e) Use the substitution $x = u^6$ to find $\int \frac{dx}{x^{\frac{1}{2}} - x^{\frac{1}{3}}}.$

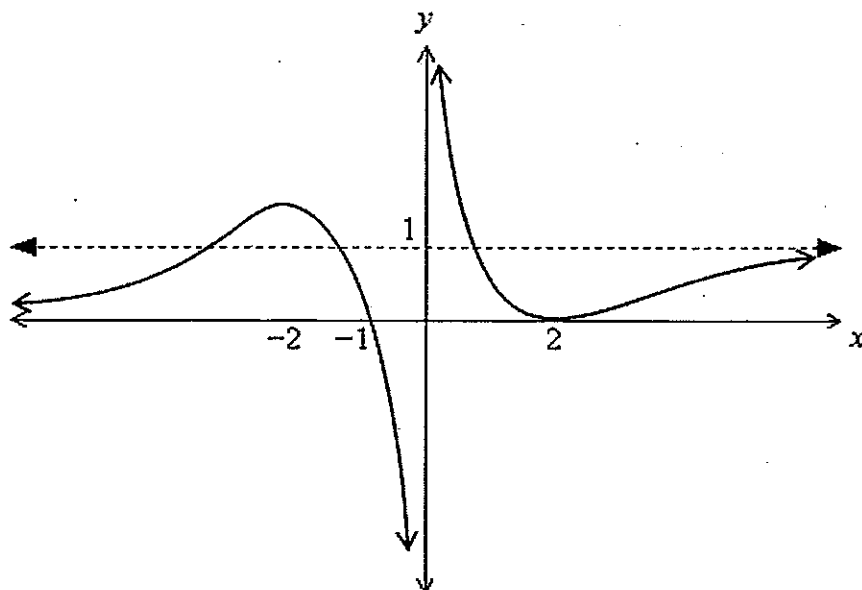
3**End of Question 2**

Question 3 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) The diagram below is a sketch of the function $y = f(x)$.
The lines $x = 0$, $y = 0$ and $y = 1$ are asymptotes.



Using the answer sheets provided, sketch each of the graphs below.

In each case, clearly label any maxima or minima, intercepts and the equations of any asymptotes.

- | | | |
|-------|-----------------------|---|
| (i) | $y = f(x)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $y^2 = f(x)$ | 2 |
| (iv) | $y = [f(x)]^2$ | 2 |
| (v) | $y = \sin^{-1}[f(x)]$ | 2 |

Question 3 continues on page 6

Question 3 (continued)

Marks

(b) Consider the curve $y = \frac{x^3 + 4}{x^2}$.

- | | | |
|-------|---|---|
| (i) | Find the coordinates of the stationary point and show that this curve is always concave up. | 2 |
| (ii) | Find the equations of any asymptotes. | 1 |
| (iii) | Sketch the curve. | 2 |
| (iv) | Find the values of k for which the equation $x^3 - kx^2 + 4 = 0$ has 3 distinct real roots. | 1 |

End of Question 3

Question 4 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) Given that $(x+1)$ is a factor of the polynomial $P(x) = x^3 + 2x^2 + 2x + 1$,
factorise $P(x)$ over the field of complex numbers . 2
- (b) The polynomial equation $x^3 - 3x^2 + 5x - 1 = 0$ has roots α, β and γ .
- (i) Find the value of $\alpha^2 + \beta^2 + \gamma^2$. 2
- (ii) Hence explain why only one root of the equation is real. 1
- (c) ω and ω^2 are the two complex cube roots of unity. 3
If ω and ω^2 are also the roots of the equation $x^3 + px^2 + qx + r = 0$,
show that $p = q$.
- (d) Given $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ and using the substitution $x = \cos \theta$,
- (i) Solve $8x^3 - 6x + 1 = 0$. 2
- (ii) Hence prove that $\sec \frac{2\pi}{9} + \sec \frac{4\pi}{9} + \sec \frac{8\pi}{9} = 6$. 2
- (e) Let the roots of $x^3 - x - 1 = 0$ be α, β and γ . 3
Find the polynomial whose roots are $\frac{1+\alpha}{1-\alpha}$, $\frac{1+\beta}{1-\beta}$ and $\frac{1+\gamma}{1-\gamma}$.

Question 5 (15 marks)

Use a SEPARATE writing booklet

Marks

(a) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$

(i) Find the coordinates of the foci and the equations of the directrices of the ellipse. 2

(ii) Sketch the ellipse, showing all key features, including intercepts. 2

(b) $P\left(2p, \frac{2}{p}\right)$ and $Q\left(2q, \frac{2}{q}\right)$ are points on the rectangular hyperbola $xy = 4$.

P and Q move on the hyperbola so that PQ always passes through $(6, 4)$.

(i) Show that $pq = \frac{p+q-3}{2}$. 3

(ii) If M is the midpoint of PQ , find the equation of the locus of M . 2

(c) The tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$) at $P(a \cos \theta, b \sin \theta)$ passes through a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricity e .

(i) Show that the tangent to the ellipse at P has equation $bx \cos \theta + ay \sin \theta = ab$. 2

(ii) Show that P lies on the directrix of the hyperbola. 2

(iii) Find the possible values of the gradient of the tangent at P . 2

Question 6 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) On the Argand diagram $P(z)$ is a point in the first quadrant of the circle $|z| = 3$.

If $\arg(z) = \theta$, find in terms of θ , expressions for:

- (i) $\arg z^4$ 1
(ii) $\arg(z-3)$. 2

- (b) A stone is projected from a point on the ground and it just clears a fence d metres away. The height of the fence is h metres. The angle of projection to the horizontal is θ and the speed of projection is v m/s. The displacement equations, measured from the point of projection are

$$x = vt \cos \theta \quad \text{and} \quad y = \frac{-1}{2} gt^2 + vt \sin \theta.$$

- (i) Show that $v^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$. 2
(ii) Show that the maximum height reached by the stone is 3

$$\frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}.$$

- (iii) Show that the stone will just clear the fence at its highest point if 3

$$\tan \theta = \frac{2h}{d}.$$

- (c) (i) If $I_n = \int_0^{\frac{1}{2}} \frac{x^n}{1-x^2} dx$ for $n = 0, 1, 2, 3, \dots$ show that 2

$$I_{n-2} - I_n = \frac{1}{(n-1)2^{n-1}} \quad \text{for } n = 2, 3, 4, \dots$$

- (ii) Given that $\int_0^{\frac{1}{2}} \frac{1}{1-x^2} dx = \frac{1}{2} \log_e 3$, find the exact value of 2

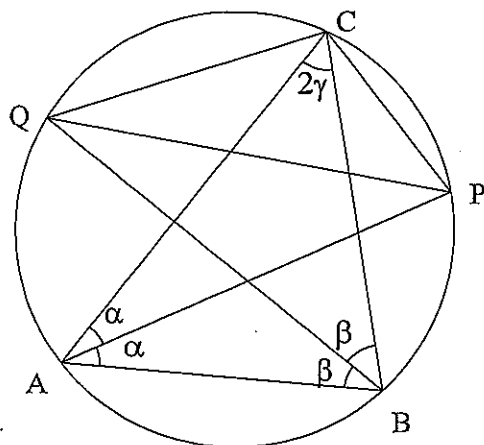
$$\int_0^{\frac{1}{2}} \frac{x^2}{1-x^2} dx.$$

Question 7 (15 marks)

Use a SEPARATE writing booklet

Marks

(a)



In the diagram above, AB is a **fixed** chord of a circle and C is a **variable** point on the major arc AB .

The angle bisectors of $\angle CAB$ and $\angle ABC$ meet the circle again at P and Q respectively.

Let $\angle CAB = 2\alpha$, $\angle ABC = 2\beta$ and $\angle BCA = 2\gamma$.

- (i) Show that $\angle PCQ = \alpha + \beta + 2\gamma$. 1
 - (ii) Hence explain why the distance PQ is constant. 2
 - (iii) Use the sine rule to show that $\frac{AB}{PQ} = 2 \sin \gamma$. 2
- (b) (i) Use DeMoivre's Theorem to show that when n is a positive integer, 2
- $$(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta} \quad (\cos \theta \neq 0).$$
- (ii) Hence show that for the equation $(1 + z)^4 + (1 - z)^4 = 0$ 2
- (where $\operatorname{Re}(z) = 0$) the roots are
- $$z = \pm i \tan \frac{\pi}{8} \text{ and } z = \pm i \tan \frac{3\pi}{8}.$$

Question 7 continues on page 11

- (c) (i) Given that $\sin x \geq \frac{2x}{\pi}$ for $0 < x < \frac{\pi}{2}$, explain why

$$\int_0^{\frac{\pi}{2}} e^{-\sin x} dx < \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx . \quad 2$$

- (ii) Show that $\int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx = \int_0^{\frac{\pi}{2}} e^{-\sin x} dx$. 2

- (iii) Hence, show that $\int_0^{\pi} e^{-\sin x} dx < \frac{\pi}{e}(e-1)$. 2

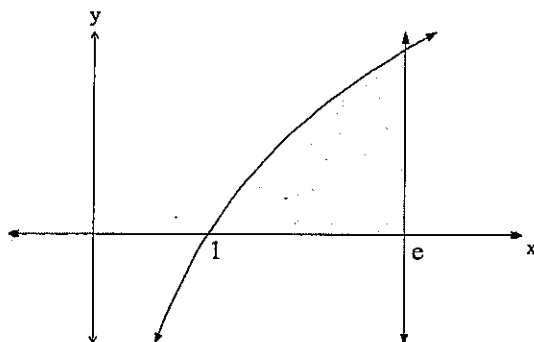
End of Question 7

Question 8 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) The diagram below shows the area bounded by the curve $y = \log_e x$, the x -axis and the line $x = e$.



This area is rotated about the y -axis to form a solid. By considering slices perpendicular to the y -axis, find the volume of the solid of revolution formed.

2

- (b) (i) Show that $\tan^{-1}\left(\frac{x}{x+1}\right) + \tan^{-1}\left(\frac{1}{2x+1}\right)$ is a constant for $2x+1 > 0$.

2

- (ii) Hence, find the exact value of the constant.

1

- (c) (i) Prove that $\frac{\cos \theta - \cos(\theta + 2\alpha)}{2\sin \alpha} = \sin(\theta + \alpha)$.

2

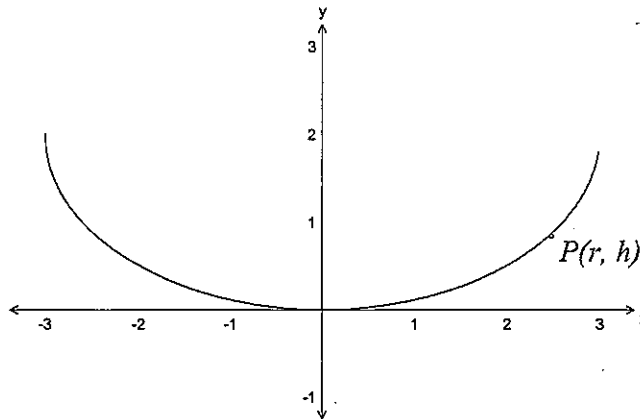
- (ii) Hence use mathematical induction to prove that

3

$$\sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta = \frac{1 - \cos 2n\theta}{2\sin \theta}.$$

Question 8 continues on page 13

- (d) The semi-ellipse given by $\frac{x^2}{9} + \frac{(y-2)^2}{4} = 1$ where $0 \leq y \leq 2$ is shown below:



The point (r, h) lies on the ellipse where $r > 0$ and $0 < h < 2$.

The tangent at P makes an angle α with the positive direction of the x -axis.

- (i) Show that $\tan \alpha = \frac{4r}{9(2-h)}$. 2
- (ii) Hence, show that $\tan \alpha = \frac{2\sqrt{4-(2-h)^2}}{3(2-h)}$. 2
- (iii) Show that the acute angle between the normal at the point P and the vertical line $x = r$ is equal to the angle between the tangent at P and the positive direction of the x -axis. 1

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

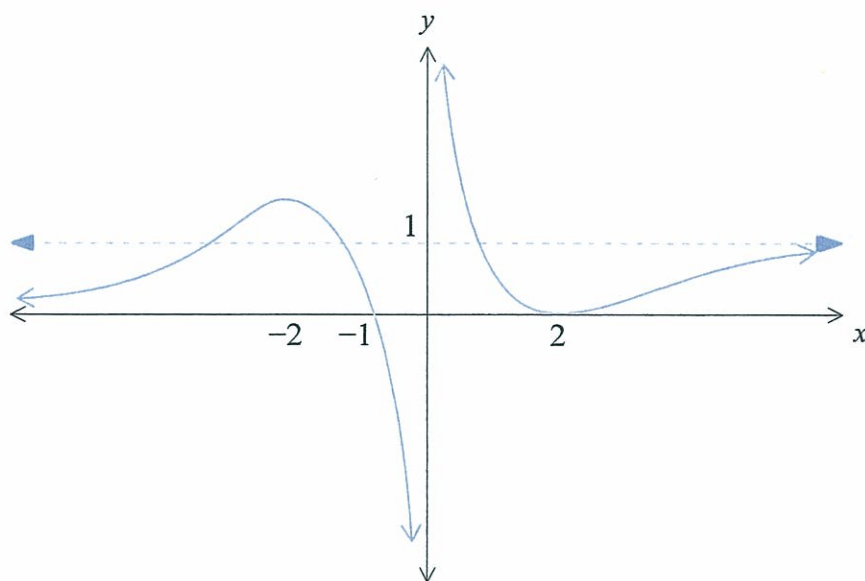
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

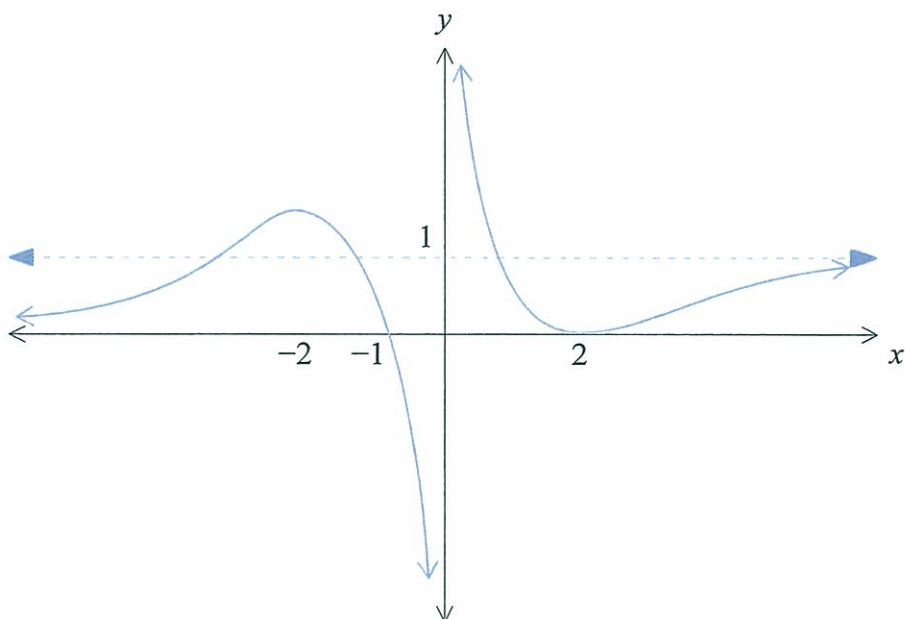
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

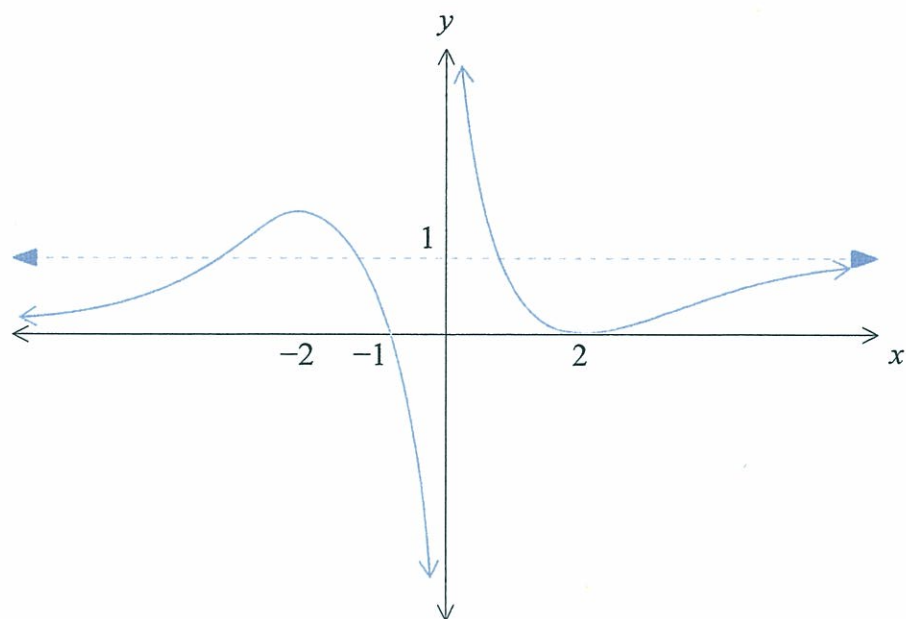
(i) $y = f(|x|)$



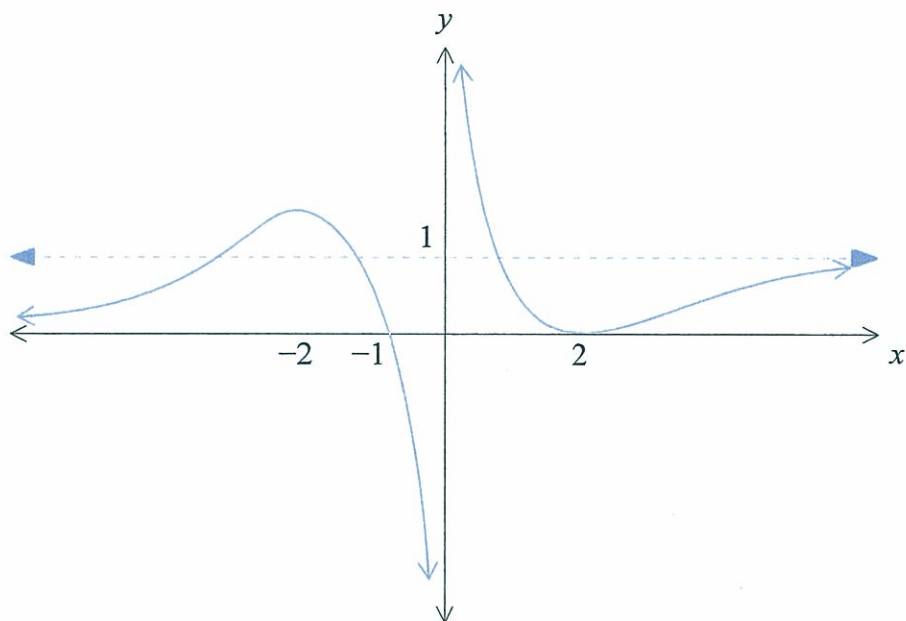
(ii) $y = \frac{1}{f(x)}$



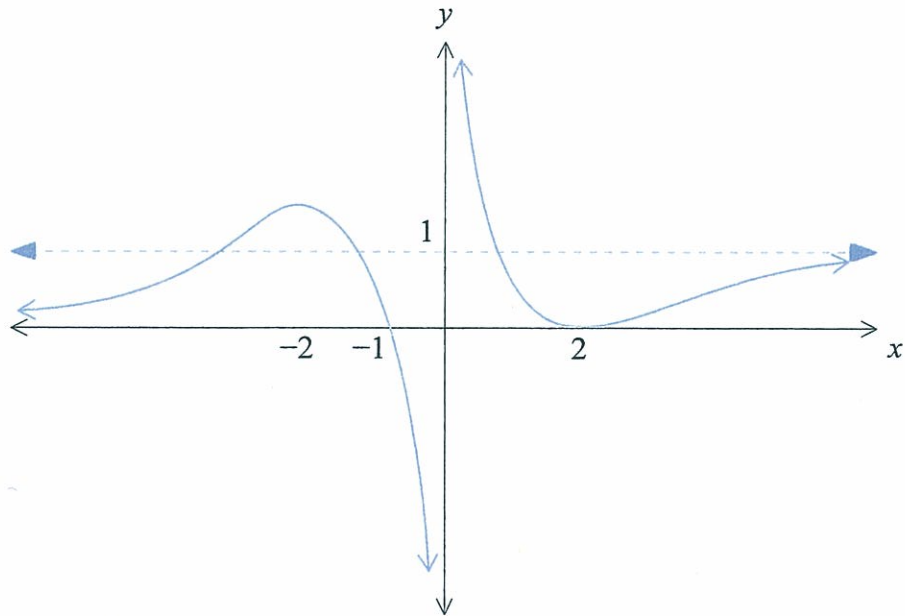
(iii) $y^2 = f(x)$



(iv) $y = [f(x)]^2$



(v) $y = \sin^{-1}[f(x)]$



QUESTION 1

(a) (i) $2z - iw$
 $= 6 + 4i - i(1+i)$
 $= 6 + 4i - i - 1$
 $= 7 + 3i$

1 R/W

(ii) $z\bar{w} = (3+2i)(1-i)$
 $= 3 - 3i + 2i + 2$
 $= 5 - i$

1 R/W

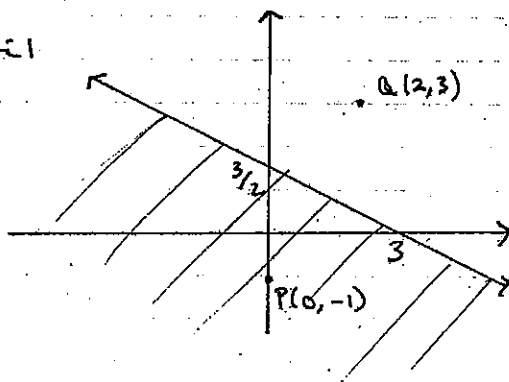
(iii) $\frac{z}{w} = \frac{3-i}{1-i}$
 $= \frac{3(1-i)}{(1-i)^2}$
 $= \frac{3-3i}{2}$

1 R/W

(b) $|z-2-3i| \geq |z+i|$

$M_{PA} = (1, 1)$
 $m_{PA} = 2$

$\therefore$ perp bisector
 $y-1 = -\frac{1}{2}(x-1)$
 $2y-2 = -x+1$
 $x+2y-3=0$



2m - all correct
 1m - correct line, incorrect shading

(c) $\alpha = -\sqrt{3} + i$

(i) $r = \sqrt{3+1} = 2$

$\theta = \frac{5\pi}{6}$

$\therefore \alpha = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

2m - both r and θ correct
 1m - one of r or θ correct
 0m - both incorrect

(ii) $z^6 + 64 = 0$

Subst α in LHS

$= 2^6\left(\cos\frac{5\pi}{6}\right)^6 + 64$

1 R/W

$= 64(\cos 5\pi) + 64$ (De Moivre)

$= 64(-1) + 64$

$= 0 \quad \therefore \alpha$ is a root of $z^6 + 64 = 0$

(iii) Since α a root of $z^6 + 64 = 0$ and $z^6 + 64$ has real coefficients then $\bar{\alpha}$ also a root.

$\therefore (x-\alpha)(x-\bar{\alpha})$ is a factor of $z^6 + 64$

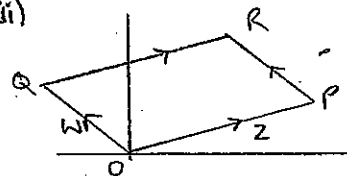
$= (x^2 - (\alpha+\bar{\alpha})x + \alpha\bar{\alpha})$

$= (x^2 - 2\operatorname{Re}(\alpha)x + |\alpha|^2)$

$= x^2 + 2\sqrt{3}x + 4$ is real quadratic factor.

(d) (i) $PA = |w-z|$ 1 R/W

(ii)



OPRQ is parallelogram.

(iii) If $|z+w| = |z-w|$ diagonals equal

$\therefore$ OPRQ is rectangle

$\therefore$ OP and OQ perpendicular

$\therefore w = kiz$ ($k = \text{const}$) and $\frac{w}{z} = \frac{kiz}{z} = ki \therefore$ imaginary.

2m - all correct including $\alpha, \bar{\alpha}$
 1m - used other factor (ie not "hence") or incorrect expansion of $(x-\alpha)(x-\bar{\alpha})$

Q2

$$\text{a) } a(2x+3) + b(x-1) = 1$$

$$\text{When } x=1, 5a = 1$$

$$a = \frac{1}{5}$$

$$\text{When } x=0, a = \frac{1}{5};$$

$$\frac{3}{5} + (-b) = 1$$

$$b = -\frac{2}{5}$$

(2)

$$\text{ii) } \int \frac{dx}{(x-1)(2x+3)}$$

$$= \frac{1}{5} \int \left(\frac{1}{x-1} - \frac{2}{2x+3} \right) dx$$

$$= \frac{1}{5} [\ln|x-1| - \ln|2x+3|] + C$$

$$= \frac{1}{5} \ln \left| \frac{x-1}{2x+3} \right| + C$$

(2)

$$\text{b) } \int \operatorname{cosec} 2x \, dx$$

$$\rightarrow t = \tan x$$

$$= \int \frac{1}{\sin 2x} \, dx$$

$$dt = \sec^2 x \, dx$$

$$dt = (\tan^2 x + 1) dx$$

$$= \int \frac{1+t^2}{2t} \cdot \frac{dt}{t^2+1} \quad \frac{dt}{t^2+1} = dx$$

$$= \frac{1}{2} \ln|t| + C$$

$$= \frac{1}{2} \ln|\tan x| + C$$

(2)

$$\text{c) } \int_{-\frac{\pi}{2}}^0 \frac{dx}{2+4x+4x^2}$$

$$= \int_{-\frac{\pi}{2}}^0 \frac{dx}{(2x+1)^2 + 1}$$

$$= \frac{1}{2} \tan^{-1}(2x+1) \Big|_{-\frac{\pi}{2}}^0$$

$$= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi}{8}$$

(2)

$$\text{d) } \int (x \cdot 3^x) \, dx$$

$$= \frac{x \cdot 3^x}{\ln 3} - \int \frac{3^x}{\ln 3} \, dx$$

$$= \frac{1}{\ln 3} \left(x \cdot 3^x - \frac{3^x}{\ln 3} \right) + C$$

$$= \frac{3^x}{\ln 3} \left(x - \frac{1}{\ln 3} \right) + C$$

(3)

$$\text{e) } \int \frac{dx}{x^{\frac{1}{2}} - x^{\frac{1}{3}}}$$

$$x = u^6$$

$$= \int \frac{6u^5}{u^3 - u^2} \, du \quad dx = 6u^5 \, du$$

$$= 6 \int \frac{u^3}{u-1} \, du$$

$$= 6 \int \left(u^2 + u + 1 + \frac{1}{u-1} \right) \, du$$

$$= 2u^3 + 3u^2 + 6u + 6 \ln|u-1| + C$$

$$= 2x^{\frac{1}{2}} + 3x^{\frac{1}{3}} + 6x^{\frac{1}{6}} + 6 \ln|x^{\frac{1}{6}} - 1| + C$$

(2)

$$y = 3^x$$

$$x \ln 3 = \ln y$$

$$x = \frac{\ln y}{\ln 3}$$

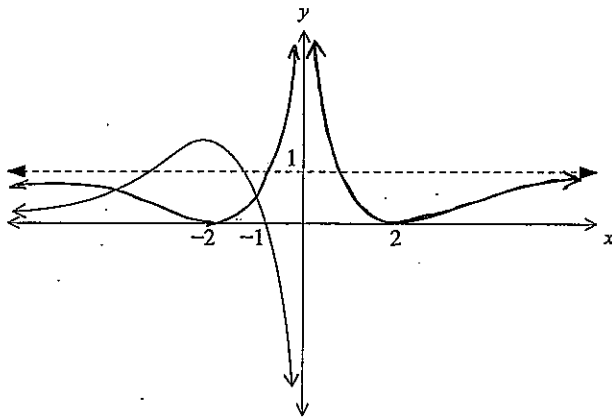
$$\frac{dx}{dy} = \frac{1}{y \ln 3}$$

$$\frac{dy}{dx} = 3^x \cdot \ln 3$$

NAME: _____

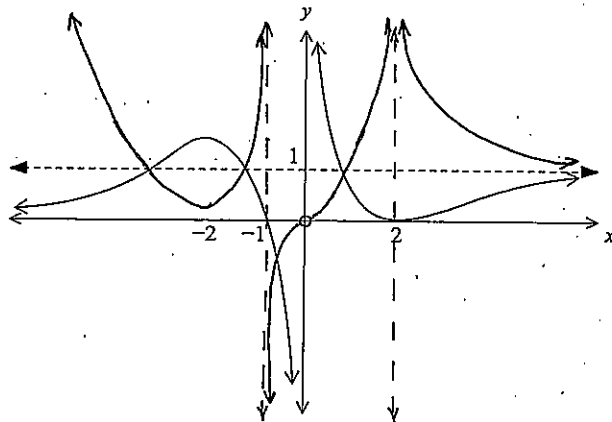
ANSWER SHEET FOR QUESTION 2(a)

(ii) Given $y = f(x)$, sketch $y = f(|x|)$.



①

(ii) Given $y = f(x)$, sketch $y = \frac{1}{f(x)}$.

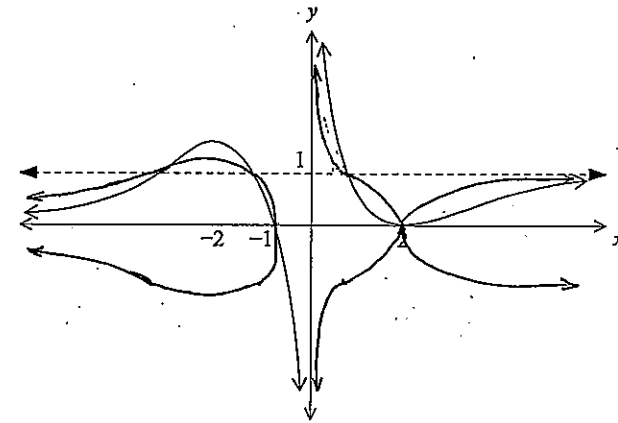


②

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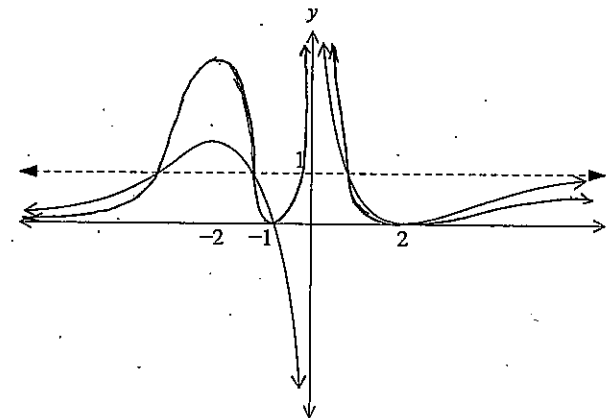
NAME: _____

(iii) Given $y = f(x)$, sketch $y^2 = f(x)$.



②

(iv) Given $y = f(x)$, sketch $y = [f(x)]^2$.

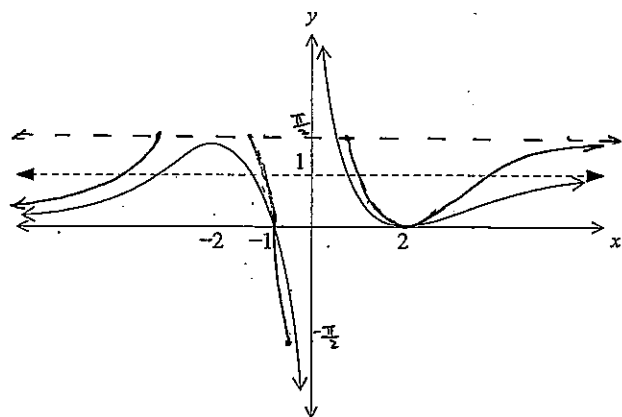


②

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NAME: _____

(v) Given $y = f(x)$, sketch $y = \sin^{-1}[f(x)]$.



②

$$b) y = \frac{x^3 + 4}{x^2}, \quad x \neq 0$$

$$= x + 4x^{-2}$$

$$i) y' = 1 - 8x^{-3}$$

$$y' = 1 - \frac{8}{x^3} = 0$$

$$x^3 = 8$$

$$x = 2$$

$$y = \frac{8+4}{4} = 3$$

Stationary pt. = (2, 3)

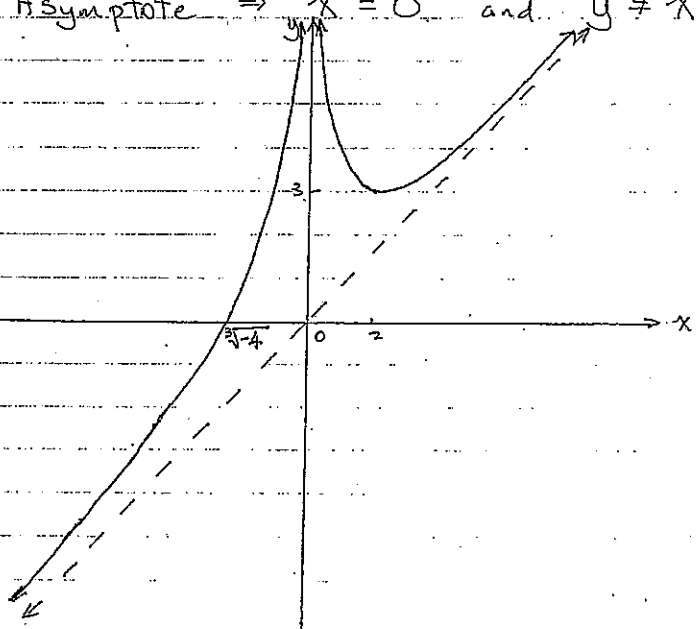
$$y'' = 24x^{-4}$$

Since $x^{-4} > 0$ for all values of x , $x \neq 0$,

$y'' > 0$ hence curve is always concave up. ③

ii) Asymptote $\Rightarrow x = 0$ and $y \neq x$ ①

iii)



1 For right asymptote and turning pt
1 for shape and 2 intercept

②

iv) $k > 3$

①

Question 4

$$P(x) = x^3 + 2x^2 + 2x + 1$$

Since $x+1$ is a factor

$$P(x) = (x+1)(x^2 + x + 1) \quad (\text{by inspection})$$

$$\text{Now zeros of } x^2 + x + 1 \quad x = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$x = \frac{-1 \pm \sqrt{3}i}{2} = \frac{-1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$\therefore P(x) = (x+1)\left(x + \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\left(x + \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$x^3 - 3x^2 + 5x - 1 = 0 \quad \text{has roots } \alpha, \beta, \gamma$$

$$\begin{aligned} (i) \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \\ &= 3^2 - 2(5) \\ &= -1 \end{aligned}$$

(ii) Since sum of squares of roots < 0 , at least one must be complex

Since coefficients real, complex roots occur in conjugate pairs.

Since polynomial degree 3, only 3 roots + 2 must be complex and therefore only one real root.

$$\text{If } w \text{ root, } w^3 + pw^2 + qw + r = 0 \quad (1)$$

$$\text{If } w^2 \text{ root, } w^6 + pw^4 + qw^2 + r = 0 \quad (2)$$

$$\text{From (1), } 1 + pw^2 + qw + r = 0$$

$$\text{In (2), } 1 + pw + qw^2 + r = 0$$

$$\text{Subtracting } w^2(p-q) - w(p-q) = 0$$

(i) (cont)

$$(w^2 - w)(p - q) = 0$$

-1

$$\therefore w^2 - w = 0$$

$$w(w-1) = 0$$

$$\text{or } p-q = 0$$

$$w = 0 \quad (\text{not poss})$$

$$\text{or } w = 1 \quad (\text{not poss since } -1)$$

$$w \text{ complex}$$

$$\therefore p = q$$

$$p = q$$

$$(d) \quad \cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$1. (i) \quad 8\cos^3\theta - 6\cos\theta + 1 = 0$$

$$\text{Subst } x = \cos\theta,$$

$$8\cos^3\theta - 6\cos\theta + 1 = 0$$

$$2(4\cos^3\theta - 3\cos\theta) = -1$$

$$4\cos^3\theta - 3\cos\theta = -1/2$$

$$\cos 3\theta = -1/2$$

$$\therefore 3\theta = \pm 2\pi/3 + 2n\pi \quad 3\theta = \pi + \pi/3, \pi + \pi/3$$

$$\theta = \pm 2\pi/9 + 2n\pi/3$$

$$3 \text{ unique solutions for } \cos\theta: \cos 2\pi/9, \cos 8\pi/9, \cos 14\pi/9$$

$$\therefore x = \cos 2\pi/9, \cos 8\pi/9, \cos 14\pi/9$$

$$= \cos 2\pi/9, \cos 8\pi/9, \cos 4\pi/9$$

$$(ii) \quad \text{Now } \cos 14\pi/9 = \cos(2\pi - 4\pi/9) = \cos 4\pi/9$$

$$\text{Let roots be } \alpha, \beta, \gamma \text{ then } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{c/a}{-d/a}$$

$$= \frac{c}{-d}$$

$$= 6$$

$$\frac{1}{\cos 2\pi/9} + \frac{1}{\cos 4\pi/9} + \frac{1}{\cos 8\pi/9} = 6$$

$$\sec 2\pi/9 + \sec 4\pi/9 + \sec 8\pi/9 = 6$$

2) $x^3 - x - 1 = 0$ Roots α, β and γ .

Require eq with roots $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$

Let $y = \frac{1+\alpha}{1-\alpha}$

$$y(1-\alpha) = (1+\alpha)$$

$$y-1 = \alpha + \alpha y$$

$$y-1 = \alpha(1+y)$$

$$\alpha = \frac{y-1}{1+y} \quad -1$$

Eq becomes

$$\left(\frac{y-1}{1+y}\right)^3 - \left(\frac{y-1}{1+y}\right) - 1 = 0$$

$$(y-1)^3 - (y-1)(1+y)^2 - (1+y)^3 = 0 \quad -1$$

$$(y^3 - 3y^2 + 3y - 1) - (y-1)(1+2y+y^2) - (y^3 + 3y^2 + 3y + 1) = 0$$

$$(y^3 - 3y^2 + 3y - 1) - (y^3 + 2y^2 + y - 1 - 2y - y^2) - (y^3 + 3y^2 + 3y + 1) = 0$$

$$-y^3 - 7y^2 + y - 1 = 0 \quad -1$$

$$y^3 + 7y^2 - y + 1 = 0$$

2m - all correct

1m - correct exp for

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

incorrect subst

Question 5

(a) $\frac{x^2}{9} + \frac{y^2}{25} = 1$

(i) $a^2 = 9 \quad b^2 = 25$

$$r = 25(1 - e^2)$$

$$e^2 = 16/25$$

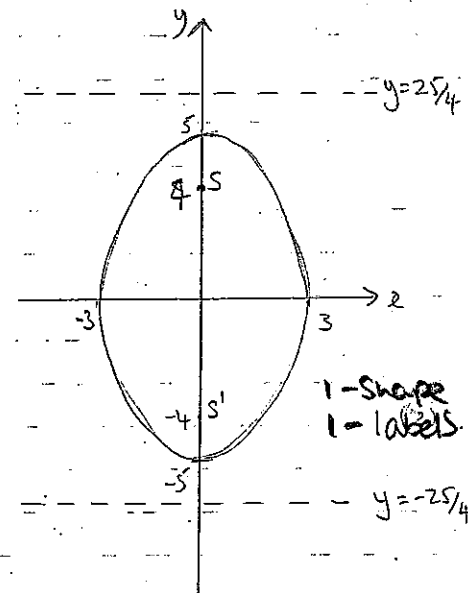
$$e = 4/5 \quad - (1)$$

foci $(0, \pm be)$

$$(0, \pm 4)$$

directrices $y = \pm \frac{b}{e}$

$$= \pm \frac{25}{4} \quad - (1)$$



(b) $P(2p, 2/p) \quad Q(2q, 2/q) \quad xy = 4$

$$m_{PQ} = \frac{2/q - 2/p}{2(q-p)}$$

$$= \frac{2p - 2q}{2pq(q-p)}$$

$$= -\frac{1}{pq}$$

Eq of PQ: $y - 2/p = -\frac{1}{pq}(x - 2p)$

Since passes through $(6, 4)$

$$4 - 2/p = -\frac{1}{pq}(6 - 2p)$$

$$4pq - 2q = -6 + 2p$$

cont.

$$2pq - q = -3 + 3p$$

$$2pq = p + q - 3$$

$$pq = \frac{p+q-3}{2} \quad \text{as required} \quad \textcircled{1}$$

$$(ii) \quad M = \left(\frac{2p+2q}{2}, \frac{\frac{2}{p} + \frac{2}{q}}{2} \right)$$

$$= (p+q, \frac{1}{p} + \frac{1}{q})$$

$$= (p+q, \frac{p+q}{pq})$$

$$\therefore x = p+q \quad \textcircled{2} \quad y = \frac{p+q}{pq} \quad \textcircled{3}$$

$\therefore$ Subst $\textcircled{2}$ into $\textcircled{3}$,

$$y = \frac{x}{pq}$$

Subst $\textcircled{1}$ into

$$y = \frac{x}{\frac{p+q-3}{2}}$$

$$y = \frac{x}{\frac{x-3}{2}}$$

$$y = \frac{2x}{x-3}$$

3m - all correct
2m - correct gradient line
1m - correct gradient.

2m - all correct

1m - if midpoint + correct
Subst of x/y with incorrect
Subst of p/q

Question 5 (cont)

$$(C) (i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{at } P(a \cos \theta, b \sin \theta)$$

$$x = a \cos \theta \quad y = b \sin \theta$$

$$\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{-b \cos \theta}{a \sin \theta}$$

Eq. of tangent,

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$a y \sin \theta + b x \cos \theta = a b (\sin^2 \theta + \cos^2 \theta)$$

$$a y \sin \theta + b x \cos \theta = a b \quad \text{//}$$

(ii) Since tangent passes through foci $(ae, 0)$ or $(-ae, 0)$

$$\text{then } \pm a b e \cos \theta = a b$$

$$e \cos \theta = \pm 1$$

$$\cos \theta = \pm \frac{1}{e} \quad \textcircled{1}$$

$$\text{Now at } P \quad x = a \cos \theta$$

$$\text{Sub. in } \textcircled{1} \quad = \pm a \cdot \frac{1}{e}$$

$$= \pm \frac{a}{e}$$

Hence P lies on directrix of hyperbola.

$$(iii) \quad \text{At } P, \quad b^2 = a^2(e^2 - 1) \quad \text{and } e \cos \theta = \pm 1 \quad \textcircled{2} \quad \textcircled{3}$$

$$\text{and } \frac{dy}{dx} = -\frac{b}{a} \frac{\cos \theta}{\sin \theta}$$

ignored
 $\pm$ error

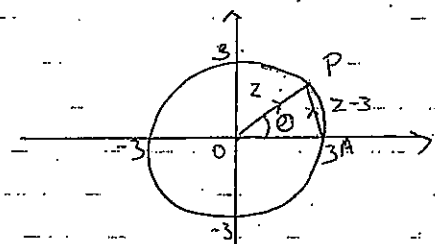
Q. (iii) cont

$$\begin{aligned}
 \left(\frac{dy}{dx}\right)^2 &= \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta} \\
 &= (e^2 - 1) \frac{\cos^2 \theta}{\sin^2 \theta} \quad (\text{from (2)}) \\
 &= \frac{e^2 \cos^2 \theta - \cos^2 \theta}{1 - \cos^2 \theta} \\
 &= \frac{1 - \cos^2 \theta}{1 - \cos^2 \theta} \quad (\text{from (3)}) \\
 &= 1 \quad \text{ie } (e \cos \theta = \pm 1) \\
 \therefore \frac{dy}{dx} &= \pm 1.
 \end{aligned}$$

Hence tangent to ellipse at P has gradient ± 1 .

2m - Correctly using
 $e \cos \theta = \pm 1$ and
 $b^2 = a^2(e^2 - 1)$
 to obtain gradients

1m - Sub either (2) +
 (3) into grad
 expression +
 finding grad
 in terms of e, a
 or b +
 not simplifying
 to ± 1 .



(i) $\arg z^4 = 4\arg z$
 $= 4\theta$

(1)

(ii) $\arg(z-3)$

$\angle OPA = \frac{\pi - \theta}{2}$ (Sum of isosceles Δ)

$\arg(z-3) = \pi - \left(\frac{\pi - \theta}{2}\right)$ (Angles on a straight line = π)

$= \frac{\pi + \theta}{2}$

$= \frac{\pi + \theta}{2}$

(2)

b) $x = vt \cos \theta$

$y = \frac{1}{2}gt^2 + vt \sin \theta$

ii) When $x = d$, $y = h$:

$d = vt \cos \theta \Rightarrow t = \frac{d}{v \cos \theta}$ — (1)

$h = \frac{1}{2}gt^2 + vt \sin \theta$ — (2)

Sub (1) into (2):

$h = \frac{-gd^2}{2v^2 \cos^2 \theta} + \frac{d v \sin \theta}{v \cos \theta}$

$2v^2 h = -gd^2 \sec^2 \theta + 2d \tan \theta v^2$

$2v^2(h - d \tan \theta) = -gd^2 \sec^2 \theta$

$2v^2(d \tan \theta - h) = gd^2 \sec^2 \theta$

$v^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$ — (3)

iii) $y = -gt + v \sin \theta$

Max. height when $y = 0$:

$gt = v \sin \theta$

$t = \frac{v \sin \theta}{g}$

Max height, y

$= \frac{-g}{2} \frac{v^2 \sin^2 \theta}{g^2} + \frac{v^2 \sin^2 \theta}{g}$

$= \frac{v^2 \sin^2 \theta}{2g}$

$= \frac{\sin^2 \theta}{2g} \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$ (from (3))

$= \frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$

$\frac{\sin^2 \theta \sec^2 \theta}{\sin^2 \theta \cdot \frac{1}{\cos^2 \theta}}$
 $= \tan^2 \theta$ — (3)

iii) Stone clears the fence at its height pt. $\Rightarrow y = h$

$4h(d \tan \theta - h) = d^2 \tan^2 \theta$

$d^2 \tan^2 \theta - 4hd \tan \theta + 4h^2 = 0$

$(d \tan \theta - 2h)^2 = 0$

$d \tan \theta = 2h$

$\tan \theta = \frac{2h}{d}$ — (3)

Q6

$$c) \text{ i) } I_n = \int_0^{\frac{1}{2}} \frac{x^n}{1-x^2} dx$$

$$I_{n-2} = \int_0^{\frac{1}{2}} \frac{x^{n-2}}{1-x^2} dx$$

$$I_{n-2} - I_n$$

$$= \int_0^{\frac{1}{2}} \frac{x^{n-2} - x^n}{1-x^2} dx$$

$$= \int_0^{\frac{1}{2}} \frac{x^{n-2} (1-x^2)}{1-x^2} dx$$

$$= \int_0^{\frac{1}{2}} x^{n-2} dx$$

$$= \frac{x^{n-1}}{n-1} \Big|_0^{\frac{1}{2}}$$

$$= \frac{1}{n-1} \left[\frac{1}{2}^{n-1} - 0 \right]$$

$$= \frac{1}{(n-1) 2^{n-1}}$$

(2)

$$\text{ii) } \int_0^{\frac{1}{2}} \frac{x^2}{1-x^2} dx = I_2$$

$$= I_{2-2} = \frac{1}{(2-1) \cdot 2^{2-1}}$$

$$= I_0 = \frac{1}{2}$$

$$= \int_0^{\frac{1}{2}} \frac{x^0}{1-x^2} dx = \frac{1}{2}$$

$$= \int_0^{\frac{1}{2}} \frac{1}{1-x^2} dx = \frac{1}{2}$$

$$= \frac{1}{2} \log_e 3 = \frac{1}{2}$$

$$= \frac{1}{2} (\ln 3 - 1)$$

(2)

Question 7

(i) $\angle PCQ = \angle PCB + \angle BCA + \angle ACQ$ (adjacent angles).

Now $\angle PCB = \angle PAB$ (angles at the circumference subtended by same arc (PB) are equal)
 $= \alpha$ (given)

Similarly, $\angle ACQ = \angle ABC$ (angles at the circumference subtended by same arc (AC) are equal)
 $= \beta$ (given)

And $\angle BCA = 2\gamma$ (given)

↑
R/W
(correct reasoning)

$\therefore \angle PCQ = \alpha + \beta + 2\gamma$

(ii) AB is fixed $\therefore$ angle subtended by AB is fixed.
 $\therefore 2\gamma = \text{constant}$
 $\therefore \gamma = \text{constant}$

In $\triangle ABC$, $2\alpha + 2\beta + 2\gamma = 180^\circ$ (angle sum of $\triangle = 180^\circ$)
 $\therefore \alpha + \beta + \gamma = 90$

Im - showing $\alpha + \beta$ const $\therefore \alpha + \beta = 90 - \gamma$
 $= \text{constant}$ since, γ constant

Im - correct conclusion.

Since $\alpha + \beta$ and 2γ are constants
 $\alpha + \beta + 2\gamma = \text{constant}$

$\therefore PQ$ is constant since subtends a fixed angle at the circumference $\angle PCQ = \alpha + \beta + 2\gamma$.

Consider $\triangle APB$, $\angle APB = 2\gamma$ (angles at circumference subtended by same arc are equal)

and $\frac{\sin 2\gamma}{AB} = \frac{\sin \alpha}{BP}$ (i) (sine rule)

Question 7 (cont)

(a) (iii) cont

Consider $\triangle QPB$, $\frac{\sin(\alpha + \beta)}{PQ} = \frac{\sin \alpha}{BP}$ (2)

From (1) and (2),

$\frac{\sin(\alpha + \beta)}{PQ} = \frac{\sin 2\gamma}{AB}$

Im - for expressions for AB and PQ using sine rule

$\frac{AB}{PQ} = \frac{\sin 2\gamma}{\sin(\alpha + \beta)}$

Im - combining + substit $\alpha + \beta = 90 - \gamma$

$= \frac{2\sin \gamma \cos \gamma}{\sin(90 - \gamma)}$ (from ii)

$= \frac{2\sin \gamma \cos \gamma}{\cos \gamma}$
 $= 2\sin \gamma$

(b) (i) LHS = $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n$

$= \left(1 + i \frac{\sin \theta}{\cos \theta}\right)^n + \left(1 - i \frac{\sin \theta}{\cos \theta}\right)^n$

$= \left(\frac{1}{\cos \theta} (\cos \theta + i \sin \theta)\right)^n + \left(\frac{1}{\cos \theta} (\cos \theta - i \sin \theta)\right)^n$

$= \frac{1}{\cos^n \theta} [(\cos \theta + i \sin \theta)^n + (\cos \theta - i \sin \theta)^n]$

$= \frac{1}{\cos^n \theta} [\cos^n \theta + i \sin^n \theta + \cos^n \theta - i \sin^n \theta]$ (De Moivre)

$= \frac{1}{\cos^n \theta} 2\cos^n \theta$

$= \text{RHS}$ 2m - must apply De Moivre correctly

shor 7 (cont)

ii) let $z = itan\theta$

$$\text{From (i)} \quad (1+z)^n + (1-z)^n = \frac{2\cos n\theta}{\cos^n \theta} = 0$$

$$\therefore \frac{2\cos n\theta}{\cos^n \theta} = 0$$

$$4 - (1+z)^4 + (1-z)^4 = 0$$

$$\frac{2\cos 4\theta}{\cos^4 \theta} = 0$$

$$\cos 4\theta = 0$$

$$4\theta = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$$

$$\theta = \pm \pi/8, \pm 3\pi/8, \pm 5\pi/8, \dots$$

$$4 \text{ unique solutions } z = \pm i \tan \pi/8, \pm i \tan 3\pi/8$$

Correct
general
soln

123

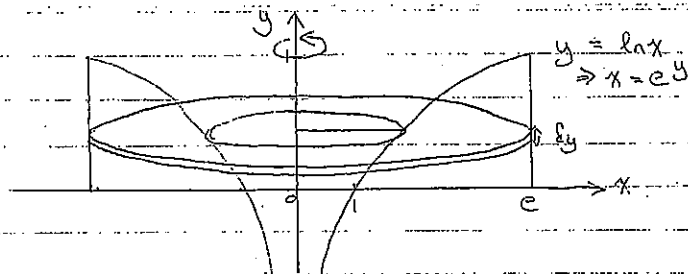
7c) i) Since e^{-x} is a decreasing function and
as $\sin x \geq \frac{2x}{\pi}$ for $0 \leq x \leq \frac{\pi}{2}$,
 $\therefore e^{-\sin x} \leq e^{-\frac{2x}{\pi}}$ for $0 \leq x \leq \frac{\pi}{2}$,
 $\therefore \int_0^{\pi/2} e^{-\sin x} dx < \int_0^{\pi/2} e^{-\frac{2x}{\pi}} dx$

$$\begin{aligned} \text{ii) } \int_{\pi/2}^{\pi} e^{-\sin x} dx & \quad \text{let } u = \pi - x \\ & = \int_{\pi/2}^0 (-e^{-\sin(\pi-u)}) du \quad du = -dx \\ & \quad x = \pi, u = \pi - \pi = 0 \\ & \quad x = \pi/2, u = \pi - \pi/2 = \pi/2 \\ & = \int_0^{\pi/2} e^{-\sin u} du \\ & = \int_0^{\pi/2} e^{-\sin x} dx \end{aligned}$$

$$\begin{aligned} \text{iii) } \int_0^{\pi} e^{-\sin x} dx & = \int_0^{\pi/2} e^{-\sin x} dx + \int_{\pi/2}^{\pi} e^{-\sin x} dx \\ & = 2 \int_0^{\pi/2} e^{-\sin x} dx \\ & < 2 \int_0^{\pi/2} e^{-\frac{2x}{\pi}} dx \\ & = 2 \left[-\frac{\pi}{2} e^{-\frac{2x}{\pi}} \right]_0^{\pi/2} \\ & = -\pi (e^{-1} - e^0) \\ & = -\pi (e^{-1} - 1) \\ & = \pi (1 - \frac{1}{e}) \\ & = \frac{\pi}{e} (e - 1) \end{aligned}$$

Q8

a)



$$\text{Area} = \pi (e^2 - x^2)$$

$$\delta V = \pi (e^2 - e^{2y}) \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^1 \pi (e^2 - e^{2y}) \delta y$$

$$= \int_0^1 \pi (e^2 - e^{2y}) dy$$

$$= \pi [e^2 y - \frac{1}{2} e^{2y}]_0^1$$

$$= \pi (e^2 - \frac{1}{2} e^2 - 0 + \frac{1}{2} e^0)$$

$$= \pi (\frac{1}{2} e^2 + \frac{1}{2})$$

$$= \frac{\pi}{2} (e^2 + 1) \text{ cubic units}$$

(2)

Q8

$$b) i) \frac{d}{dx} \tan^{-1} \left(\frac{x}{x+1} \right)$$

$$= \frac{1}{1 + \left(\frac{x}{x+1} \right)^2} \cdot \frac{1(x+1) - 1 \cdot x}{(x+1)^2}$$

$$= \frac{(x+1)^2}{(x+1)^2 + x^2} \cdot \frac{1}{(x+1)^2}$$

$$= \frac{1}{2x^2 + 2x + 1}$$

$$\frac{d}{dx} \tan^{-1} \left(\frac{1}{2x+1} \right)$$

$$= \frac{1}{1 + \left(\frac{1}{2x+1} \right)^2} \cdot \frac{-2}{(2x+1)^2}$$

$$= \frac{(2x+1)^2}{(2x+1)^2 + 1} \cdot \frac{-2}{(2x+1)^2}$$

$$= \frac{-2}{4x^2 + 4x + 2}$$

$$= \frac{-1}{2x^2 + 2x + 1}$$

$$\frac{d}{dx} \left[\tan^{-1} \left(\frac{x}{x+1} \right) + \tan^{-1} \left(\frac{1}{2x+1} \right) \right]$$

$$= \frac{1}{2x^2 + 2x + 1} - \frac{1}{2x^2 + 2x + 1}$$

$$= 0$$

$$\therefore \tan^{-1} \left(\frac{x}{x+1} \right) + \tan^{-1} \left(\frac{1}{2x+1} \right) \text{ is a constant.} \quad (2)$$

$$ii) \text{ When } x = 0,$$

$$\tan^{-1} (0) + \tan^{-1} (1)$$

$$= 0 + \frac{\pi}{4}$$

$$= \frac{\pi}{4}$$

$$\therefore \text{Constant} = \frac{\pi}{4}$$

(1)

$$\begin{aligned}
 \text{c2 ii) LHS} &= \frac{\cos \theta - \cos(\theta + 2\alpha)}{2 \sin \alpha} \\
 &= \frac{\cos \theta - \cos \theta \cos 2\alpha + \sin \theta \sin 2\alpha}{2 \sin \alpha} \\
 &= \frac{\cos \theta - \cos \theta (1 - 2 \sin^2 \alpha) + 2 \sin \theta \sin \alpha \cos \alpha}{2 \sin \alpha} \\
 &= \frac{\cos \theta - \cos \theta + 2 \sin^2 \alpha \cos \theta + 2 \sin \theta \sin \alpha \cos \alpha}{2 \sin \alpha} \\
 &= \sin \alpha \cos \theta + \sin \theta \cos \alpha \\
 &= \sin(\theta + \alpha) \\
 &= \text{RHS}
 \end{aligned}$$

iii) let the statement be

$$\sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta = \frac{1 - \cos 2n\theta}{2 \sin \theta}$$

Step 1: Test the statement is true for $n=1$.

$$\text{LHS} = \sin[2(1)-1]\theta$$

$$= \sin \theta$$

$$\text{RHS} = \frac{1 - \cos 2\theta}{2 \sin \theta}$$

$$= \frac{1 - (1 - 2 \sin^2 \theta)}{2 \sin \theta}$$

$$= \sin \theta$$

$$= \text{LHS}$$

$$= \text{LHS}$$

(3)

Step 2: Assume statement is true for $n=k$;

$$\text{i.e. } \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2k-1)\theta$$

$$= \frac{1 - \cos 2k\theta}{2 \sin \theta}$$

Step 3: Prove statement is true for $n=k+1$;

$$\text{i.e. prove } \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2k+1)\theta$$

$$= \frac{1 - \cos(2k+2)\theta}{2 \sin \theta}$$

$$\text{LHS} = \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2k-1)\theta + \sin(2k+1)\theta$$

$$= \frac{1 - \cos 2k\theta}{2 \sin \theta} + \sin(2k+1)\theta$$

$$= \frac{1 - \cos 2k\theta}{2 \sin \theta} + \sin(2k\theta + \theta)$$

$$= \frac{1 - \cos 2k\theta}{2 \sin \theta} + \frac{\cos 2k\theta - \cos(2k\theta + 2\theta)}{2 \sin \theta} \quad \text{from (i)}$$

$$= \frac{1 - \cos 2k\theta + \cos 2k\theta - \cos(2k+2)\theta}{2 \sin \theta}$$

$$= \frac{1 - \cos(2k+2)\theta}{2 \sin \theta}$$

$$= \text{RHS}$$

Step 4: Since the statement is true for $n=1$,

assumed true for $n=k$ and proved true for $n=k+1$,

by mathematical induction it is true for $n=1+1=2$,

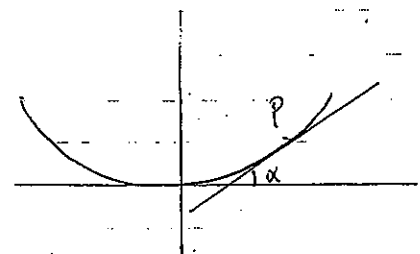
$n=2+1=3$, and so on. Hence the statement is

true for all positive integers of n .

1 For Steps 2 and 4

(3)

Question 8 (cont)



$$\frac{x^2}{9} + \frac{(y-2)^2}{4} = 1$$

Differentiating

$$\frac{2x}{9} + \frac{2(y-2)}{4} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{9} \times \frac{2}{y-2}$$

$$= \frac{4x}{9(2-y)}$$

At P, $m_p = \frac{4r}{9(2-h)}$

$m_p = \tan \alpha$

$\therefore \tan \alpha = \frac{4r}{9(2-h)}$

2

i) (r, h) lies on ellipse,

$$\frac{r^2}{9} + \frac{(h-2)^2}{4} = 1$$

$$\frac{r^2}{9} = 1 - \frac{(h-2)^2}{4}$$

$$= \frac{4 - (h-2)^2}{4}$$

$$r^2 = 9 \left(\frac{4 - (h-2)^2}{4} \right) \quad (h-2)^2 = (2-h)^2$$

$$r = \frac{3\sqrt{4 - (2-h)^2}}{2} \quad r > 0$$

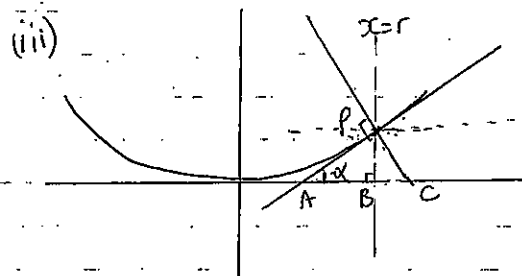
Question 8c (cont)

$$\tan \alpha = \frac{4}{9(2-h)} \times \frac{3\sqrt{4 - (2-h)^2}}{2}$$

$$= \frac{2\sqrt{4 - (2-h)^2}}{3(2-h)}$$

as required.

2



$\angle APB = 90^\circ - \alpha$ (angle sum of $\triangle APB = 180^\circ$)

$\angle APC = 90^\circ$ (angle between tangent and normal)

$\angle CPB = \angle APC - \angle APB$

$= 90^\circ - (90^\circ - \alpha)$

$= \alpha$

$\therefore$ acute angle between normal and $x=r$, $\angle CPB = \alpha$
as required.

1